

Frequency Bias Signal Malfunctions

Symptoms

- Engine does **not** respond to frequency bias request.

How To Use This Tree

This symptom tree can be used to troubleshoot engine symptoms. Start by performing Step 1 troubleshooting. Step 2 will ask a series of questions and will provide a list of troubleshooting steps to perform, depending on the symptom.

Shoptalk

None.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check the customer interface box (C.I.B.) wiring.	
	STEP 1A. Check the generator speed/load governing bias SUPPLY wire.	Less than 10 ohms?
	STEP 1B. Check the generator speed/load governing bias RETURN wire.	Less than 10 ohms?
	STEP 1C. Check the generator speed/load governing bias SIGNAL wire.	Less than 10 ohms?
STEP 2.	Check the engine harness to the C.I.B.	
	STEP 2A. Check the generator speed/load governing bias SUPPLY and SIGNAL wires.	Less than 10 ohms?

STEPS		SPECIFICATIONS
	STEP 2B. Check the generator speed/load governing bias RETURN and SIGNAL wires.	Less than 10 ohms?

STEP 1. Check the customer interface box (C.I.B.) wiring.

STEP 1A. Check the generator speed/load governing bias SUPPLY wire.

Conditions :

- Open the C.I.B.
- Disconnect the C.I.B. to the engine harness cable connector C1 from the C.I.B.

Action	Specification/Repair	Next Step
Check the generator speed/load governing bias SUPPLY wire. • Place one test lead on the generator speed/load governing bias 5 volt SUPPLY pin in connector C1. • Place the other test lead on the generator speed/load governing bias 5 volt SUPPLY terminal on the X1 connector.	Less than 10 ohms? YES	1B
	Less than 10 ohms? NO Repair : Replace the wire. Refer to Procedure 015-023 in Section 15. (/qs3/pubsys2/xml/en/procedures/300/300-015-023.html)	Repair complete

STEP 1B. Check the generator speed/load governing bias RETURN wire.

Conditions :

- Open the C.I.B.
- Disconnect the C.I.B. to the engine harness cable connector C1 from the C.I.B.

Action	Specification/Repair	Next Step
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Check the generator speed/load governing bias RETURN wire. - Place one test lead on the generator speed/load governing bias RETURN pin in connector C1. - Place the other test lead on the generator speed/load governing bias return terminal on the X1 connector.	Less than 10 ohms? YES	2A
	Less than 10 ohms? NO Repair : Replace the wire. Refer to Procedure 015-023 in Section 15. (/qs3/pubsys2/xml/en/procedures/300/300-015-023.html)	Repair complete

STEP 1C. Check the generator speed/load governing bias SIGNAL wire.

Conditions : - Open the C.I.B. - Disconnect the C.I.B. to the engine harness cable connector C1 from the C.I.B.		
Action	Specification/Repair	Next Step
Check the generator speed/load governing bias SIGNAL wire. - Place one test lead on the generator speed/load governing bias SIGNAL pin in connector C1. - Place the other test lead on the generator speed/load governing bias signal terminal on the X1 connector.	Less than 10 ohms? YES	2A
	Less than 10 ohms? NO Repair : Replace the wire. Refer to Procedure 015-023 in Section 15. (/qs3/pubsys2/xml/en/procedures/300/300-015-023.html)	Repair complete

STEP 2. Check the engine harness to the C.I.B.

STEP 2A. Check the generator speed/load governing bias SUPPLY and SIGNAL wires.

Conditions :

- Disconnect the C.I.B. to the engine harness cable connector C1 from the C.I.B.
- Disconnect the C.I.B. to the engine harness cable from the engine.

Action	Specification/Repair	Next Step
Check the generator speed/load governing bias SUPPLY and SIGNAL wires. • Place a jumper between the generator speed/load governing bias 5 volt SUPPLY pin and the frequency bias SIGNAL pin in the engine-side connection. • Place one test lead in the generator speed/load governing bias 5 volt SUPPLY pin of the C1 connector. • Place the other test lead in the generator speed/load governing bias SIGNAL pin of the C1 connector.	Less than 10 ohms? YES	2B
	Less than 10 ohms? NO Repair : Replace the cable.	Repair complete

STEP 2B. Check the generator speed/load governing bias RETURN and SIGNAL wires.**Conditions :**

- Disconnect the C.I.B. to the engine harness cable connector C1 from the C.I.B.
- Disconnect the C.I.B. to the engine harness cable from the engine.

Action	Specification/Repair	Next Step

Check the generator speed/load governing bias SUPPLY and SIGNAL wires. - Place a jumper between the frequency bias RETURN pin and the generator speed/load governing bias SIGNAL pin in the engine-side connection. - Place one test lead in the generator speed/load governing bias RETURN pin of the C1 connector. - Place the other test lead in the frequency bias SIGNAL pin of the C1 connector.	Less than 10 ohms? YES Repair : Use the following for potentiometer repair instructions. Reference the Marine Auxiliary QSB7-DM CM850 Fault Code Troubleshooting Manual, Bulletin 4325972, Section TF; the ISM and QSM11 Electronic Control System Troubleshooting and Repair Manual, Bulletin 3666266, Section TF; X15 CM2350 X125M Fault Code Troubleshooting Manual, Bulletin 5504346, Section TF; or the equipment manufacturer service information.	Repair complete
	Less than 10 ohms? NO Repair : Replace the cable.	Repair complete